Joint Modeling of Longitudinal and Survival Data via Second-Order ODEs

A Mechanistic Approach to Biomarker Dynamics and Event Risk

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Motivating Example: PSA Velocity in Prostate Cancer[1]

Goal: Joint modeling of biomarker trajectories and survival outcomes.

PSA Kinetics - Patient A

- Baseline PSA: 4.0 ng/mL
- PSA velocity: 0.35 ng/mL/year
- Pattern: Stable rise
- Low risk progression

PSA Kinetics - Patient B

- Baseline PSA: 4.0 ng/mL
- PSA velocity: 2.0 ng/mL/year
- Pattern: Rapid rise
- High risk progression

Can we leverage biomarker velocity to improve risk prediction?

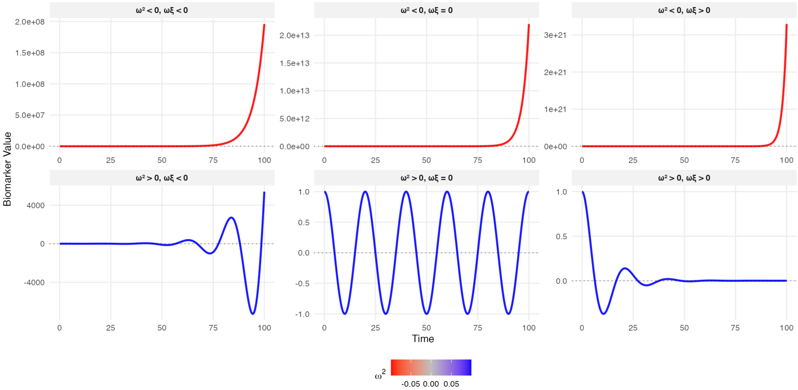
[1] D'Amico AV, et al. JAMA. 2005;294(4):440-447.

Our Solution: Mechanistic Modeling via ODEs

Second-order ODE framework for biomarker dynamics:

$$\boxed{\ddot{m}(t) + 2\xi\omega\dot{m}(t) + \omega^2 m(t) = k\omega^2 u(t)}$$

ξ: damping ratio, ω: natural frequency, k: gain, $u(t)$: excitation



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Stability Conditions:

- **Stable:** $\omega^2 > 0$ and $\omega\xi > 0$
- **Unstable:** $\omega^2 \leq 0$ or $\omega\xi \leq 0$

Can we test $H_0 : \omega^2 \leq 0$ or $\omega\xi \leq 0$ vs $H_1 : \omega^2 > 0$ and $\omega\xi > 0$?

Damping Effects

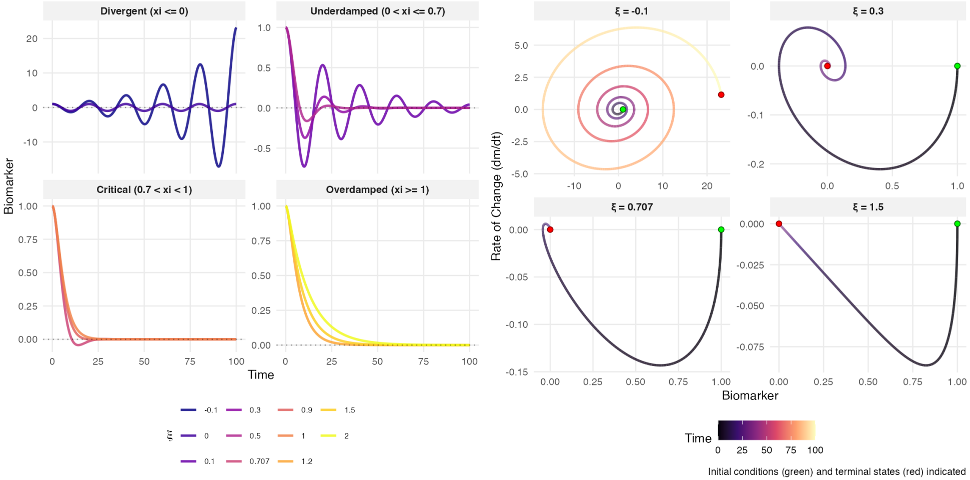


Figure: Autonomous systems with no external excitation $(u(t) = 0)$

Damping Effects (cont.)

The damping ratio ξ determines convergence behavior

Damping (ξ)	Regime	Behavior
$\xi \leq 0$	Unstable	Divergence or sustained oscillations
$0 < \xi \leq 0.7$	Underdamped	Decaying oscillations
$0.7 < \xi < 1$	Near-critical	Rapid convergence, minimal overshoot
$\xi \geq 1$	Overdamped	Slow monotonic decay

Special cases:

- $\xi = \frac{\sqrt{2}}{2} \approx 0.707$: Fastest settling time with overshoot
- $\xi = 1$: Critical damping, the fastest response without oscillation

External Interventions

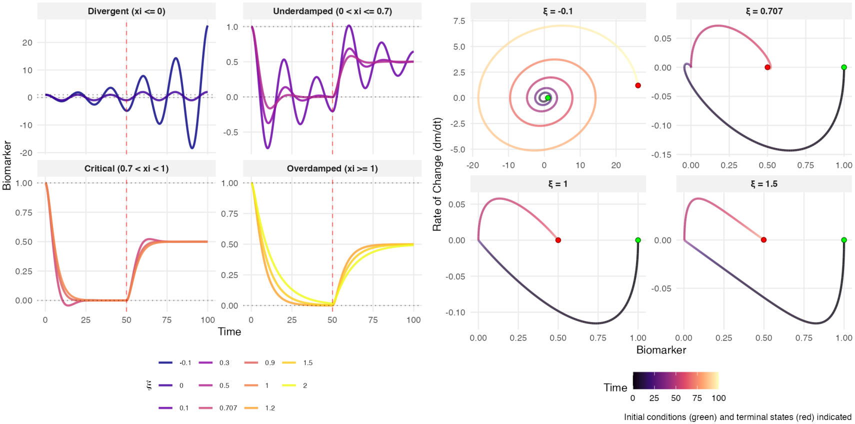


Figure: Non-autonomous systems with step excitation $(u(t) = 0.5 \cdot \mathrm{I}\{t \geq 50\})$

Biological Trajectories Naturally Follow ODE Dynamics

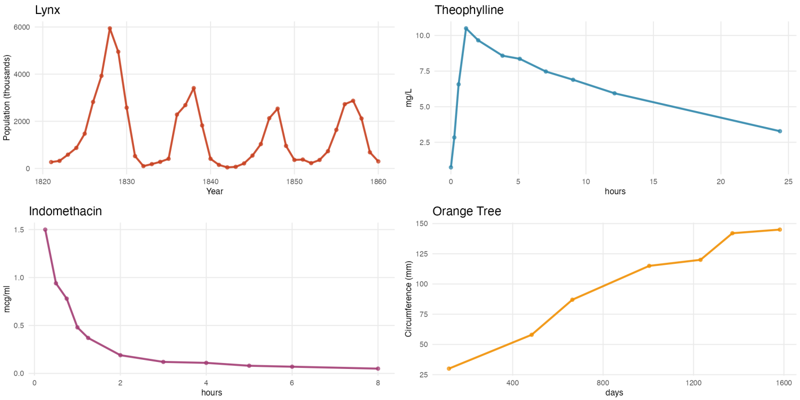


Figure: Diverse biological systems naturally exhibit second-order ODE dynamics. **Top-left:** Lynx population cycles. **Top-right:** Theophylline pharmacokinetics. **Bottom-left:** Indomethacin concentration. **Bottom-right:** Orange tree growth.

Joint Modeling Framework

Longitudinal Submodel:

$$V_{ij} = m_i(T_{ij}) + b_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_e^2), \quad b_i \sim \mathcal{N}(0, \sigma_b^2)$$

Biomarker dynamics governed by:

$$\ddot{m}_i(t) + 2\xi\omega\dot{m}_i(t) + \omega^2 m_i(t) = k\omega^2 u_i(t) \Rightarrow$$
$$\ddot{m}_i(t) = \beta_1 m_i(t) + \beta_2 \dot{m}_i(t) + \boldsymbol{\beta}_X^\top \boldsymbol{X}_i(t)$$

with initial conditions: $m_i(0) = m_{i0}$, $\dot{m}_i(0) = \dot{m}_{i0}$

Survival Submodel:

$$\lambda_i(t) = \lambda_0(t) \exp\left[\underbrace{\alpha_1 m_i(t)}_{\text{level}} + \underbrace{\alpha_2 \dot{m}_i^\gamma(t)}_{\text{volatility}} + \boldsymbol{W}_i(t)^\top \boldsymbol{\phi} + b_i\right]$$

where $\gamma \in \{1, 2\}$ for linear or squared velocity effects, and $\lambda_0(t)$ is approximated by B-splines $\log(\lambda_0(t)) = \boldsymbol{\eta}^\top \mathbf{B}^{(\lambda)}(t)$

Unified ODE System

Following Tang et al. (2023)[3], the survival process can be naturally integrated through the relationship $\frac{d\Lambda(t)}{dt} = \lambda(t)$.

Define the state vector: $\boldsymbol{s}_i(t) = [\Lambda_i(t), m_i(t), \dot{m}_i(t)]^\top$. The unified system evolves as:

$$\boxed{\frac{d\boldsymbol{s}_i}{dt} = \begin{pmatrix} \lambda_i(t|b_i) \\ \dot{m}_i(t) \\ \ddot{m}_i(t) \end{pmatrix}, \quad \text{with} \quad \boldsymbol{s}_i(0) = \begin{pmatrix} 0 \\ m_{i0} \\ \dot{m}_{i0} \end{pmatrix}}$$

Single ODE solver handles both longitudinal and survival processes simultaneously.

[3] Tang W, et al. J. Am. Stat. Assoc. 2023;118(544):2406-2421.

Parameter Estimation: EM Algorithm I

We employ the EM algorithm to handle the latent random effects $b_i \sim \mathcal{N}(0, \sigma_b^2)$

Parameters: $\boldsymbol{\theta} = (\omega, \xi, k, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\phi}, \boldsymbol{\eta}, \sigma_e^2, \sigma_b^2)$
Observed data: $\mathcal{O}_i = \{(V_{ij}, T_{ij})_{j=1}^{n_i}, T_i, \delta_i\}$

E-step: Given $\boldsymbol{\theta}^{(k)}$, compute posterior moments

$$\hat{b}_i^{(k)} = E[b_i|\mathcal{O}_i; \boldsymbol{\theta}^{(k)}], \quad \hat{v}_i^{(k)} = \mathrm{Var}[b_i|\mathcal{O}_i; \boldsymbol{\theta}^{(k)}]$$

Parameter Estimation: EM Algorithm (cont.)

M-step: Maximize the expected complete-data log-likelihood

$$
\begin{aligned}
Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)}) &= E[\ell_c(\boldsymbol{\theta})|\mathcal{O};\boldsymbol{\theta}^{(k)}] \\
&= \sum_{i=1}^{n} E\Big[\log f(V_i|b_i;\boldsymbol{\theta}) + \log f(T_i,\delta_i|b_i;\boldsymbol{\theta}) + \log f(b_i;\boldsymbol{\theta})\Big|\mathcal{O}_i;\boldsymbol{\theta}^{(k)}\Big]
\end{aligned}
$$

Update parameters:

$$
\boldsymbol{\theta}^{(k+1)} = \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})
$$

Iterate E and M steps until convergence.

Simulation Design

Data Generation Process

- **Sample:** $n = 100$ subjects, annual measurements per subject in $[0, 10]$ years
- **Biomarker dynamics:** Second-order ODE with parameters
  - Natural frequency: $\omega = 2\pi/5$ (period $=$ 5 years)
  - Damping ratio: $\xi \in \{0.7, 1.2\}$ (critical/overdamped)
- **Hazard model:** $\lambda_i(t) = \lambda_0(t) \exp[\alpha_1 m_i(t) + \alpha_2 \dot{m}_i(t) + \boldsymbol{\phi}^\top \mathbf{W}_i + b_i]$

Benchmark Comparison

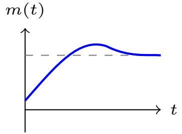
- **Proposed:** Joint ODE model estimating all parameters
- **Oracle:** Cox regression using true $m_i(t_j)$, $\dot{m}_i(t_j)$ at observation times

Simulation Results (Critical Damping: $\xi = 0.7$)

Parameter	Method	True	Bias	ESE	ASE	CP(%)
ϕ_1	Oracle	0.40	0.014	0.135	0.127	93
	Proposed		0.018	0.132	0.130	94
ϕ_2	Oracle	-0.60	-0.022	0.261	0.250	94
	Proposed		0.002	0.277	0.252	91
α_1 (level)	Oracle	0.30	0.047	0.259	0.239	92
	Proposed		**0.037**	**0.134**	**0.132**	**96**
α_2 (velocity)	Oracle	1.00	-0.067	0.643	0.585	92
	Proposed		**−0.032**	**0.218**	**0.223**	**96**
$T = 2\pi/\omega$	Proposed	5.00	0.008	0.045	0.039	91
ξ	Proposed	0.70	0.000	0.010	0.009	94

Critical Damping

$\xi = 0.7$



Oscillatory convergence

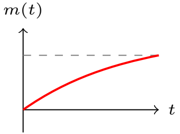
Note: ESE: empirical SE; ASE: asymptotic SE; CP: coverage probability.

Simulation Results (Overdamped: $\xi = 1.2$)

Parameter	Method	True	Bias	ESE	ASE	CP(%)
ϕ_1	Oracle	0.40	0.011	0.136	0.131	93
	Proposed		0.015	0.136	0.134	94
ϕ_2	Oracle	-0.60	-0.033	0.276	0.258	92
	Proposed		-0.023	0.280	0.260	92
α_1 (level)	Oracle	0.30	0.040	0.258	0.238	92
	Proposed		$\mathbf{0.042}$	$\mathbf{0.159}$	$\mathbf{0.151}$	**94**
α_2 (velocity)	Oracle	1.00	-0.105	1.070	0.943	90
	Proposed		$\mathbf{-0.138}$	$\mathbf{0.449}$	$\mathbf{0.412}$	**94**
$T = 2\pi/\omega$	Proposed	5.00	0.015	0.086	0.083	92
ξ	Proposed	1.20	-0.003	0.026	0.027	94

Overdamped

$\xi = 1.2$



Monotonic convergence

Note: ESE: empirical SE; ASE: asymptotic SE; CP: coverage probability.

Extrapolation Capability of Proposed Method

man/figures/README-visualization-1.png

Summary

Methodological Innovation:

1. Mechanistic modeling of biomarker trajectories via second-order ODEs
2. Natural incorporation of velocity

Future Directions:

- **Heterogeneous dynamics:** Allow patient-specific ω and ξ parameters to capture population heterogeneity
- **Variable selection:** Develop methods for selecting informative biomarkers when multiple candidates are available
- **Biomarker interactions:** Model cross-talk and feedback loops between multiple biomarkers via coupled ODEs

Computational Challenges

Individual ODE Solutions

- Each patient requires solving their own ODE system
- Computational cost scales linearly with sample size: $\mathcal{O}(n)$

Gradient Computation via Augmented ODEs

- Computing $\nabla_{\boldsymbol{\theta}}\mathcal{L}$ requires solving an augmented system:

$$\frac{d}{dt}\begin{pmatrix} \boldsymbol{s}(t) \\ \frac{\partial \boldsymbol{s}}{\partial \boldsymbol{\theta}} \end{pmatrix} = \begin{pmatrix} f(\boldsymbol{s}, \boldsymbol{\theta}, t) \\ \frac{\partial f}{\partial \boldsymbol{s}}\frac{\partial \boldsymbol{s}}{\partial \boldsymbol{\theta}} + \frac{\partial f}{\partial \boldsymbol{\theta}} \end{pmatrix}$$

- Dimension grows from d to $d(1+p)$ where $p =$ number of parameters
- Significant memory and computation overhead

Thank You

Questions & Discussion

Resources:

`github.com/ziyangg98/JointODE`

Thank you for your attention and feedback

Appendix: ODE Solution Forms and Damping Behaviors

Damping	Condition	Solution Form	Behavior
Underdamped	$0 < \xi < 1$	$e^{-\xi\omega t}[A\cos(\omega_d t) + B\sin(\omega_d t)]$	Decaying oscillations
Critical	$\xi = 1$	$(A + Bt)e^{-\omega t}$	Fastest decay
Overdamped	$\xi > 1$	$Ae^{\lambda_1 t} + Be^{\lambda_2 t}$	Slow decay
Undamped	$\xi = 0$	$A\cos(\omega t) + B\sin(\omega t)$	Sustained oscillations
Unstable	$\omega^2 < 0$ or $\omega\xi < 0$	Exponential growth	System diverges

- $\omega_d = \omega\sqrt{1-\xi^2}$: damped natural frequency
- $\lambda_{1,2} = -\xi\omega \pm \omega\sqrt{\xi^2 - 1}$: overdamped eigenvalues
- A, B: constants determined by initial conditions $m_0, \dot{m}_0$

Appendix: Model Reparameterization

We reparameterize the ODE for easier estimation:

$$\ddot{m}(t) + 2\xi\omega\dot{m}(t) + \omega^2 m(t) = k\omega^2 u(t)$$

$\Downarrow$ Reparameterization

$$\ddot{m}(t) = \beta_1 m(t) + \beta_2 \dot{m}(t) + \boldsymbol{\beta}_X^\top \mathbf{X}(t)$$

Parameter Recovery:

$$\begin{array}{lcl} \omega & = & \sqrt{-\beta_1} \\ \xi & = & -\beta_2/(2\sqrt{-\beta_1}) \\ \text{SE} & \text{via} & \text{Delta method: } \nabla g(\boldsymbol{\beta})^\top \mathrm{Var}(\hat{\boldsymbol{\beta}}) \nabla g(\boldsymbol{\beta}) \end{array}$$

Appendix: EM Algorithm

1. **Initialization:** Set $\boldsymbol{\theta}^{(0)}$, $k \leftarrow 0$
2. **Main Loop:**
   2a. *ODE phase:* For each patient i
   - Solve augmented ODE system $\rightarrow (m_i, \frac{\partial m_i}{\partial \boldsymbol{\theta}})$
   - Compute hazard components

   2b. *Posterior phase:* Compute $\hat{b}_i$, $E[e^{b_i}]$
   2c. *Gradient phase:* Evaluate $\nabla_{\boldsymbol{\theta}} Q$
   2d. *Update:* $\boldsymbol{\theta}^{(k+1)} \leftarrow$ L-BFGS-B step
3. **Convergence:** $\|\nabla Q\| < 10^{-4}$ or $|\Delta Q| < 10^{-6}$

Appendix: Posterior Computation

Key Insight - Factorization:

$$\lambda_i(t|b_i) = e^{b_i}\lambda_i(t|0), \quad \Lambda_i(t|b_i) = e^{b_i}\Lambda_i(t|0)$$

Computational Strategy:

1. Solve ODE once with $b_i = 0$ to get $\Lambda_i(t|0)$
2. Find posterior mode by maximizing:

$$\ell_i(b) = b\left[\frac{S_i}{\sigma_e^2} + \delta_i\right] - \frac{b^2}{2}\left[\frac{n_i}{\sigma_e^2} + \frac{1}{\sigma_b^2}\right] - e^b\Lambda_i(T_i|0)$$

   where $S_i = \sum_j (V_{ij} - m_i(T_{ij}))$
3. Compute moments via Gauss-Hermite quadrature

Appendix: Gradient-Based Optimization

Objective: Maximize $Q(\boldsymbol{\theta})$ where $\boldsymbol{\theta} = (\boldsymbol{\eta}, \boldsymbol{\phi}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \sigma_e^2, \sigma_b^2)$

Gradient Decomposition:

$$\nabla_{\boldsymbol{\theta}} Q = \underbrace{\sum_{i,j} \frac{r_{ij}}{\sigma_e^2} \frac{\partial m_i(T_{ij})}{\partial \boldsymbol{\theta}}}_{\text{Longitudinal component}} + \underbrace{\sum_i \left[\delta_i \frac{\partial \log \lambda_i}{\partial \boldsymbol{\theta}} - E[e^{b_i}] \frac{\partial \Lambda_i}{\partial \boldsymbol{\theta}} \right]}_{\text{Survival component}}$$

where $r_{ij} = V_{ij} - m_i(T_{ij}) - \hat{b}_i$ is the residual

No analytical form $\rightarrow$ numerical computation required

Appendix: Gradient-Based Computation cont.

Strategy: Solve state and sensitivities Simultaneously

Denote $\mathbf{Z}_i(t) = [m_i(t), \dot{m}_i(t), \mathbf{X}_i(t)^\top]^\top$, then the augmented system is:

$$\frac{d}{dt}\begin{pmatrix} \Lambda_i(t) \\ m_i(t) \\ \dot{m}_i(t) \\ \frac{\partial \Lambda_i}{\partial \boldsymbol{\beta}} \\ \frac{\partial m_i}{\partial \boldsymbol{\beta}} \\ \frac{\partial \dot{m}_i}{\partial \boldsymbol{\beta}} \end{pmatrix} = \begin{pmatrix} \lambda_i(t|0) \\ \dot{m}_i(t) \\ \boldsymbol{\beta}^\top \mathbf{Z}_i(t) \\ \frac{\partial \lambda_i}{\partial \boldsymbol{\beta}} + \lambda_i \boldsymbol{\alpha}^\top \frac{\partial \mathbf{m}_i}{\partial \boldsymbol{\beta}} \\ \frac{\partial \dot{m}_i}{\partial \boldsymbol{\beta}} \\ \mathbf{Z}_i(t) + \boldsymbol{\beta}^\top \frac{\partial \mathbf{Z}_i(t)}{\partial \boldsymbol{\beta}} \end{pmatrix}$$

Initial State $(0, m_{i0}, \dot{m}_{i0})$, Sensitivities all zero